

Overview

KO-CAP Polymer Capacitors

The KEMET Organic Capacitor (KO-CAP) is a solid electrolytic capacitor with a conductive polymer cathode capable of delivering very low ESR and improved capacitance retention at high frequencies. KO-CAP combines the low ESR of multilayer ceramic, the high capacitance of aluminum electrolytic, and the volumetric efficiency of tantalum into a single surface mount package. Unlike liquid electrolyte-based capacitors, KO-CAP has a very long operational life and high ripple current capabilities.

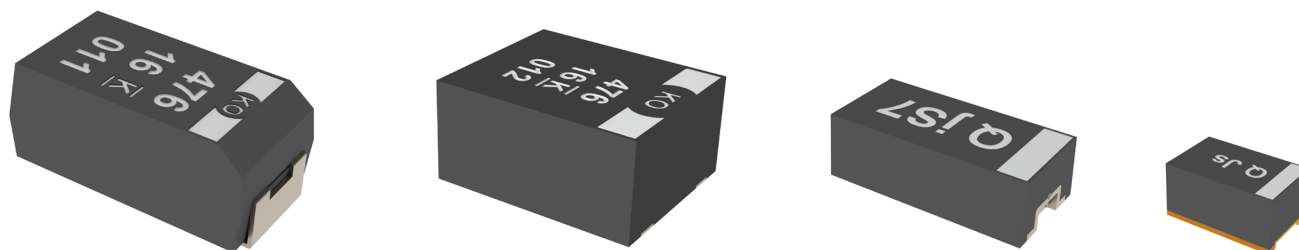
The T52X/T530 provides the widest range of voltages, capacitance and case size options in the KO-CAP family and is suitable for general purpose DC applications for up to 48 volt DC voltage rails.

Benefits

- ESR values down to 5 mΩ
- Stable capacitance across temperature and voltage
- No aging effects
- High ripple handling
- Volumetrically efficient
- High frequency capacitance retention
- 100% accelerated steady state aging
- 100% surge current tested
- Halogen-free epoxy and RoHS compliant

Applications

Typical applications include DC/DC converters, audio/sound circuits (mobile phone and base stations, smart phones, MP3 players), power supply inputs, portable electronics (notebook PCs, displays, SSDs, HDDs and USBs, digital cameras, GPS navigation systems, WiFi modules), telecommunications, consumer electronics (analytical and test equipment, high speed servers), high voltage applications such as 12 V to 48 V power input rails, densely populated circuits with space restrictions, microprocessor decoupling, and high ripple current applications.



Environmental Compliance

- RoHS compliant (6/6) according to Directive 2002/95/EC when ordered with 100% Sn, Ni-Pd-Au or non-magnetic 100% Sn solder
- Halogen-free
- Epoxy compliant with UL94 V-0

K-SIM

For a detailed analysis of specific part numbers, please visit ksim.kemet.com to access KEMET's K-SIM software. KEMET K-SIM is designed to simulate behavior of components with respect to frequency, ambient temperature, and DC bias levels.

Series Reference Selection

Series		Voltage Range		Temperature Rating			Special Features	
		< 1 V to 9 V Applications	12 V to 48 V Applications	85°C Rated	105°C Rated	125°C Rated	Miniature Size	Low ESL
T520	Standard	X	*	X**	X			
T521	High Voltage		X		X	X		
T523	Facedown Terminal	X	X**	X	X**			X
T525	High Temperature	X	*			X		
T527	Small Size	X	X		X		X	
T529	Miniature	X	X		X		X	
T530	High Cap/ Low ESR	X	X		X	X		
TF08	Ultrathin-Facedown	X	*		X		X	

* Not recommended for new design.

** Selected values.

Ordering Information

T52X/T530

T	520	V	157	M	006	A	T	E045	
Capacitor Class	Series	Case Size ¹	Capacitance Code (pF)	Capacitance Tolerance	Rated Voltage (VDC)	Failure Rate/Design	Termination Finish	ESR Code	Packaging (C-Spec)
T = Tantalum	520 = Polymer 521 = High voltage 523 = Facedown terminal 525 = 125°C rated 527 = Facedown terminal 529 = Substrate terminal 530 = 125°C High capacitance	A B C D H I J L M P Q S T U V W X Y	First two digits represent significant figures. Third digit specifies number of zeros.	K = ±10% M = ±20%	002 = 2 2R5 = 2.5 003 = 3 004 = 4 006 = 6.3 008 = 8 010 = 10 011 = 11 12R = 12.5 016 = 16 020 = 20 025 = 25 035 = 35 050 = 50 063 = 63 075 = 75	A = N/A	T = 100% Matte tin (Sn)-plated H*** = Tin/lead (SnPb) solder coated (5% Pb minimum) P* = Ni-Pd-Au-plated N = Non-magnetic 100% Tin (Sn) M = Non-magnetic (SnPb) A** = Ni-Au	E = ESR last three digits specify ESR in mΩ (045 = 45 mΩ)	Blank = 7" reel 7280**** = 13" reel

Gold termination available upon request

* P termination only available on select part numbers

** A termination only available on T529 part numbers

*** H termination not available for T527/T529 part numbers

**** 13" reel not available for T520A case and some other part numbers

Ordering Information cont.

TF08

T	F	08A	226	M	016	A	P	E200
Capacitor Class	Series	Case Size	Capacitance Code (pF)	Capacitance Tolerance	Rated Voltage (VDC)	Failure Rate/ Design	Termination Finish	ESR Code
T = Tantalum	F= Ultrathin polymer solution-Facedown construction	08A = 3216/0.8 max height	First two digits represent significant figures. Third digit specifies number of zeros.	M = ±20%	006 = 6.3 016 = 16	A = N/A	P = Ni-Pd-Au-plated	E = ESR Last three digits specify ESR in mΩ (200 = 200mΩ)

Performance Characteristics

Item	Performance Characteristics
Operating Temperature	-55°C to 85°C/105°C/125°C (refer to part number for maximum temperature rating)
Rated Capacitance Range	4.7 – 1,500 µF at 120 Hz/25°C
Capacitance Tolerance	K tolerance (10%), M tolerance (20%)
Rated Voltage Range	2 – 75 V
DF (120 Hz)	Refer to Part Number Electrical Specification Table*
ESR (100 kHz)	Refer to Part Number Electrical Specification Table**
Leakage Current	Refer to Part Number Electrical Specification Table

* For T529 Series DF measurement apply 1.5 V DC

** For PN T520B477M2R5ATE014 the ESR measurement should be taken at 300 to 500kHz

Qualification

Test	Condition	Characteristics			
Endurance	Temperature: 85°C, 105°C, 125°C* ² Voltage: 1.0 Rated Voltage 2/3 Rated Voltage for 125°C rating PN Time: 2,000 Hours 1,000 Hours for T527/T529* ³	Δ C/C	Within -20%/+10% of initial value T527, T529 and TF08A: Within -20/+20 of initial value		
		DF	Within initial limits T527/T529/TF08A: Within 1.5 x initial limits* ³		
		DCL	85°C and 105°C rated parts: Within 1.25 x initial limit 125°C rated parts: Within 2.0 x initial limits T527/TF08A: Within initial limits T529: Within 3.0 x initial limits		
		ESR	Within 2.0 x initial limit (N/A for T527/T529/TF08A* ³)		
Storage Life	Temperature: 105°C, 125°C* ² Voltage: 0 Voltage Time: 2,000 Hours (N/A for T527 and T529* ³)	Δ C/C	Within -20%/+10% of initial value		
		DF	Within initial limits		
		DCL	Within 1.25 x initial limit T521: 1.25 x IL at 105°C, 2x IL at 125°C T525/T530: Within 2.0 x initial limit		
		ESR	Within 2.0 x initial limit		
Humidity	Temperature: 60°C Voltage: No Load Humidity: 90% RH Time: 500 Hours, 1,000 Hours (T525, T530)	Δ C/C	Within -5%/+35% of initial value T527/T529/TF08A: -20% to +30% of initial DC/C limit T520B477M2R5ATE014: -20% to +50% of initial DC/C limit		
		DF	Within initial limits T527/T529/TF08A: Within 1.5 x initial limits* ³		
		DCL	Within 5.0 x initial limit T521/T525/T529/T530: Within 3.0 x initial limit* ³ T527/TF08A: Within initial limits		
		ESR	Within 2.0 x initial limit T525/T527/T529/T530/TF08A: N/A* ³		
Temperature Stability	Extreme temperature exposure at a succession of continuous steps at +25°C, -55°C, +25°C, +85°C, +105°/125°C* ² , +25°C	Δ C/C	+25°C	-55°C	+85°C
			IL* ¹	±20% -20% - 0% of DC/C (T527/29/TF08A)* ³	±20%
			DF	IL	1.2 x IL 1.5 x IL* ⁶
			DCL	IL	N/A
Surge Voltage	Temperature: 105°C, 85°C* ^{2, *4, *5} Voltage: 1.32 x Rated voltage* ⁵ Time: 1,000 cycles	Δ C/C	Within -20%/+10% of initial value T527/T529/TF08A: Within -20/+20% of initial value		
		DF	Within initial limits		
		DCL	Within initial limits		
		ESR	Within initial limits T527/T529/TF08A: N/A* ³		
Mechanical Shock/Vibration	MIL-STD-202, Method 213, Condition I, 100 G Peak. MIL-STD-202, Method 204, Condition D, 10 Hz to 2,000 Hz, 20 G peak	Δ C/C	Within ±10% of initial value Within initial DC/C limits		
		DF	Within initial limits		
		DCL	Within initial limits		

*¹ IL = Initial limit

*² Please refer to part number specifications for individual temperature classification.

*³ This condition/exclusion applies also to PN T520B477M2R5ATE014 and PN T520B337M2R5ATE009

*⁴ For Series T527, T529 and TF test temperature is 85°C

*⁵ For T527 > 20 V test temperature is at +15 - +35°C, test voltage at 1.16 x V_R

For T527/TF, 6 V test voltage is 1.27 x rated voltage / For TFA08226M016 test voltage is 1.15 x rated voltage

*⁶ For part numbers rated at 85°C

Reliability

KO-CAP capacitors have an average failure rate of 0.5 %/1,000 hours at category voltage, U_C , and category temperature, T_C . These capacitors are qualified using industry test standards at U_C and T_C . The minimum test time (1,000 hours or 2,000 hours) is dependent on the product.

The actual life expectancy of KO-CAP capacitors increases when application voltage, U_A , and application temperature, T_A , are lower than U_C and T_C . As a general guideline, when $U_A < 0.9 * U_C$ and $T_A < 85^\circ\text{C}$, the life expectancy will typically exceed the useful lifetime of most hardware (> 10 years).

The lifetime of a KO-CAP capacitor at a specific application voltage and temperature can be modeled using the equations below. A failure is defined as passing enough current to blow a 1-amp fuse. The calculation is an estimation based on empirical results and is not a guarantee.

$$VAF = \left(\frac{U_C}{U_A} \right)^n$$

where:

VAF = acceleration factor due to voltage, unitless

U_C = category voltage, volt

U_A = application voltage, volt

n = exponent, 16

$$TAF = e^{\left[\frac{E_a}{k} \left(\frac{1}{273+T_A} - \frac{1}{273+T_C} \right) \right]}$$

where:

TAF = acceleration factor due to temperature, unitless

E_a = activation energy, 1.4 eV

k = Boltzmann's constant, 8.617E-5 eV/K

T_A = application temperature, $^\circ\text{C}$

T_C = category temperature, $^\circ\text{C}$

$$AF = VAF * TAF$$

where:

AF = acceleration factor, unitless

TAF = acceleration factor due to temperature, unitless

VAF = acceleration factor due to voltage, unitless

$$Life_{U_A, T_A} = Life_{U_C, T_C} * AF$$

where:

$Life_{U_A, T_A}$ = guaranteed life application voltage and temperature, years

$Life_{U_C, T_C}$ = guaranteed life category voltage and temperature, years

AF = acceleration factor, unitless

Terms:

Category voltage, U_C : maximum recommended peak DC operating voltage for continuous operation at the category temperature, T_C

Rated voltage, U_R : maximum recommended peak DC operating voltage for continuous operation up to the rated temperature, T_R

Category temperature, T_C : maximum recommended operating temperature. Voltage derating may be required at T_C

Rated temperature, T_R : maximum recommended operating temperature without voltage derating. T_R is equal to or lower than T_C

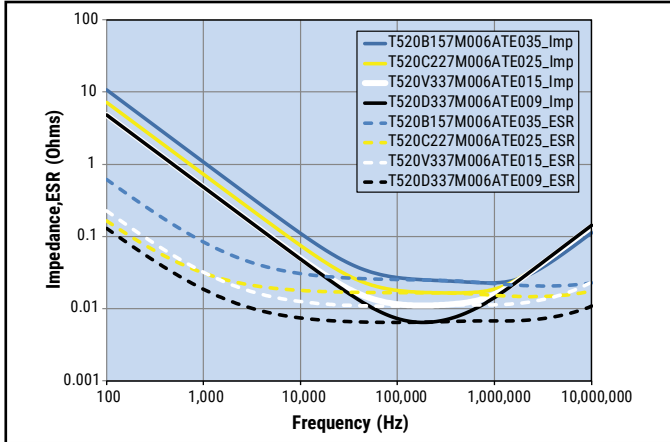
Reliability Table 1 – Common temperature range classifications

85°C (T_R) / 85°C (T_C)	Rated Voltage (U_R)	2.5	4.0	6.3	8.0	10.0	12.5	16.0	20.0	25.0	35.0	50.0	63.0	75.0
	Category Voltage (U_C)	2.5	4.0	6.3	8.0	10.0	12.5	16.0	20.0	25.0	35.0	50.0	63.0	75.0
105°C (T_R) / 105°C (T_C)	Rated Voltage (U_R)	2.5	4.0	6.3	8.0	10.0	12.5	16.0	20.0	25.0	35.0	50.0	63.0	75.0
	Category Voltage (U_C)	2.5	4.0	6.3	8.0	10.0	12.5	16.0	20.0	25.0	35.0	50.0	63.0	75.0
105°C (T_R) / 125°C (T_C)	Rated Voltage (U_R)	2.5	4.0	6.3	8.0	10.0	12.5	16.0	20.0	25.0	35.0	50.0	63.0	75.0
	Category Voltage (U_C)	1.7	2.7	4.2	5.4	6.7	8.4	10.7	13.4	16.8	23.5	33.5	42.2	50.3

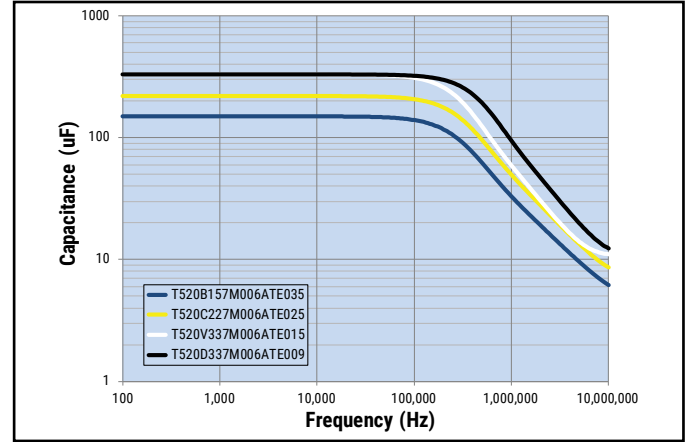
Electrical Characteristics

T520 Standard (2 V – 10 V)

ESR vs. Frequency

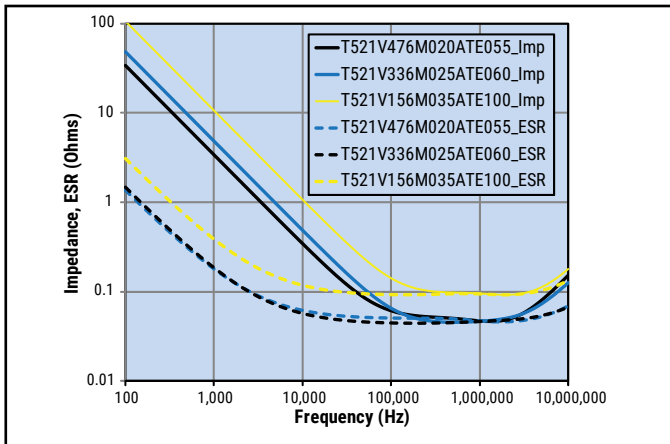


Capacitance vs. Frequency

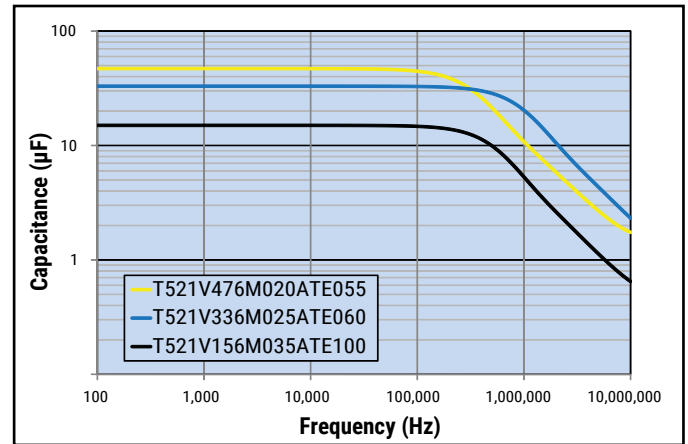


T521 High Voltage (12 V – 75 V)

ESR vs. Frequency

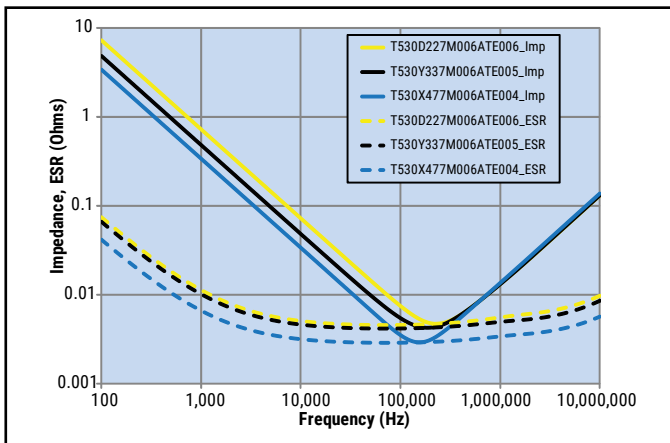


Capacitance vs. Frequency

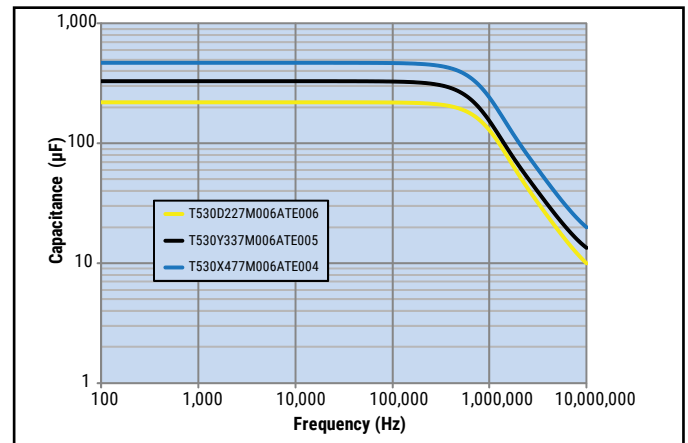


T530 High Cap/Low ESR (2 V – 10 V)

ESR vs. Frequency



Capacitance vs. Frequency

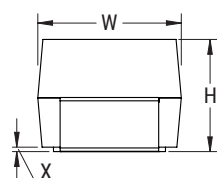


Dimensions – Millimeters (Inches)

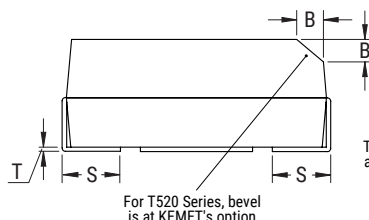
Metric will govern

T520 / T521 / T525 / T530

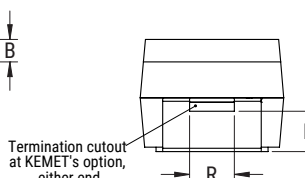
CATHODE (-) END VIEW



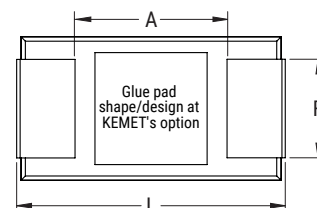
SIDE VIEW



ANODE (+) END VIEW



BOTTOM VIEW



Case Size		Component Dimensions											
KEMET	EIA	L	W	H	F	S	B ±0.15	X	P	R	T	A	Typical Weight (mg)
					±0.1 ±(0.004)	±0.3 ±(0.012) S1 ±0.4 (0.0157) S2 ±0.2 (0.00)	(Ref) ±0.006	(Ref)	(Ref)	(Ref)	(Ref)	(Min)	
A	3216-18	3.2 ±0.2 (0.126 ±0.008)	1.6 ±0.2 (0.063 ±0.008)	1.6 ±0.2 (0.063 ±0.008)	1.2 (0.047)	0.8 (0.032)	N/A	0.10 ±0.10 (0.004 ±0.004)	0.4 (0.016)	0.4 (0.016)	0.13 (0.005)	1.2 (0.047)	53
B	3528-21	3.5 ±0.2 (0.138 ±0.008)	2.8 ±0.2 (0.110 ±0.008)	1.9 ±0.1*3 (0.075 ±0.008)	2.2 (0.087)	0.8 (0.032) S1 = 0.8 (0.032)*2 S2 = 0.8 (0.032)*2	0.4 (0.016)	0.10 ±0.10 (0.004 ±0.004)	0.5 (0.020)	1.0 (0.039)	0.13 (0.005)	1.9 (0.075)	95
C	6032-28	6.0 ±0.3 (0.236 ±0.012)	3.2 ±0.2 (0.126 ±0.008)	2.5 ±0.3 (0.098 ±0.012)	2.2 (0.087)	1.30 (0.051)	0.5 (0.020)	0.10 ±0.10 (0.004 ±0.004)	0.9 (0.035)	1.0 (0.039)	0.13 (0.005)	2.9 (0.114)	184
D	7343-31	7.3 ±0.3 (0.287 ±0.012)	4.3 ±0.3 (0.169 ±0.012)	2.8 ±0.3 (0.110 ±0.012)	2.4 (0.094)	1.30 (0.051)	0.5 (0.020)	0.10 ±0.10 (0.004 ±0.004)	0.9 (0.035)	1.0 (0.039)	0.13 (0.005)	3.6 (0.142)	435
H	7360-20	7.3 ±0.3 (0.287 ±0.012)	6.0 ±0.3 (0.236 ±0.012)	1.9 ±0.1 (0.075 ±0.004)	4.1 (0.161)	1.3 (0.051)	N/A	0.10 ±0.10 (0.004 ±0.004)	N/A	N/A	0.13 (0.005)	3.3 (0.130)	385
J	7360-15	7.3 ±0.3 (0.287 ±0.012)	6.0 ±0.3 (0.236 ±0.012)	1.5 (0.059) Maximum	4.1 (0.161)	1.3 (0.051)	N/A	0.10 ±0.10 (0.004 ±0.004)	N/A	N/A	N/A	3.3 (0.130)	263
L	6032-19	6.0 ±0.3 (0.236 ±0.012)	3.2 ±0.2 (0.110 ±0.008)	1.8 ±0.1 (0.071 ±0.004)	2.2 (0.087)	1.3 (0.051)	N/A	0.05 (0.002)	N/A	N/A	0.13 (0.005)	2.5 (0.098)	187
M	3528-15	3.5 ±0.2 (0.138 ±0.008)	2.8 ±0.2 (0.110 ±0.008)	1.4 ±0.1 (0.055 ±0.004)	2.2 (0.087)	0.8 (0.031)	N/A	0.05 (0.002)	N/A	N/A	0.13 (0.005)	1.1 (0.043)	98
Q	7343-12	7.3 ±0.3 (0.287 ±0.012)	4.3 ±0.3 (0.169 ±0.012)	1.1 ±0.1 (0.043 ±0.004)	2.4 (0.094)	1.3 (0.051)	N/A	0.05 (0.002)	N/A	N/A	0.13 (0.005)	3.8 (0.150)	170
T	3528-12	3.5 ±0.2 (0.138 ±0.008)	2.8 ±0.2 (0.110 ±0.008)	1.1 ±0.1 (0.043 ±0.004)	2.2 (0.087)	0.8 (0.032)	N/A	0.05 (0.002)	N/A	N/A	0.13 (0.005)	1.9 (0.075)	55
U	6032-15	6.0 ±0.3 (0.236 ±0.012)	3.2 ±0.2 (0.126 ±0.008)	1.4 ±0.1 (0.055 ±0.004)	2.2 (0.087)	1.3 (0.051)	N/A	0.05 (0.002)	N/A	N/A	0.13 (0.005)	2.9 (0.114)	117
V	7343-19	7.3 ±0.3 (0.287 ±0.012)	4.3 ±0.3 (0.169 ±0.012)	1.8 ±0.1*1 (0.071 ±0.004)	2.4 (0.094)	1.3 (0.051)	N/A	0.05 (0.002)	N/A	N/A	0.13 (0.005)	3.6 (0.142)	274
W	7343-15	7.3 ±0.3 (0.287 ±0.012)	4.3 ±0.3 (0.169 ±0.012)	1.4 ±0.1 (0.055 ±0.004)	2.4 (0.094) 2.8 ±0.2 (0.110 ±0.0079)*2	1.30 (0.051) S1 = 5.0 (0.197)*2 S2 = 1.3 (0.051)*2	N/A	0.05 (0.002)	N/A	N/A	0.13 (0.005)	3.6 (0.142)	223
X	7343-43	7.3 ±0.3 (0.287 ±0.012)	4.3 ±0.3 (0.169 ±0.012)	4.0 ±0.3 (0.157 ±0.012)	2.4 (0.094)	1.3 (0.051)	0.5 (0.020)	0.10 ±0.10 (0.004 ±0.004)	1.7 (0.067)	1.0 (0.039)	0.13 (0.005)	3.6 (0.142)	554
Y	7343-40	7.3 ±0.3 (0.287 ±0.012)	4.3 ±0.3 (0.169 ±0.012)	3.8 ±0.2 (0.150 ±0.008)	2.4 (0.094)	1.3 (0.051)	0.5 (0.020)	0.10 ±0.10 (0.004 ±0.004)	1.7 (0.067)	1.0 (0.039)	0.13 (0.005)	3.8 (0.150)	494

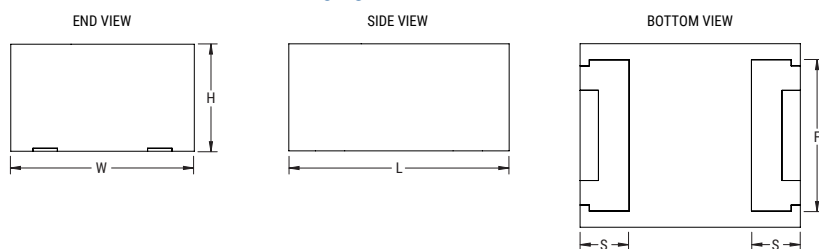
*1 Height tolerance is ±0.2 for V case T521 only

*2 S1 and S2 is for T528 series only

*3 Height tolerance is ±0.2 for: T520B with M & N terminations

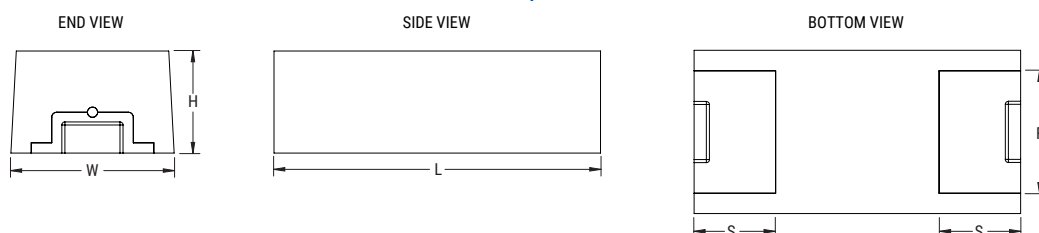
Dimensions – Millimeters cont.

T523



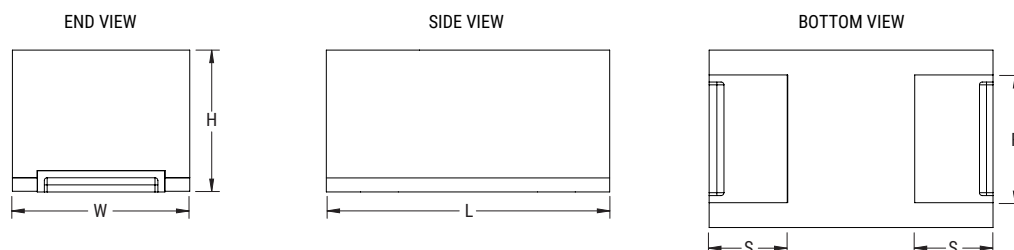
KEMET	EIA	L	W	H	F ±0.1 (±0.004)	S ±0.3 (±0.012)	Typical Weight (mg)
W	7343-15	7.3±0.3 (0.287±0.012)	4.3±0.3 (0.169±0.012)	1.4±0.1 (0.055±0.004)	2.4 (0.094)	1.3 (0.051)	223
J	7360-15	7.3±0.3 (0.287±0.012)	6.0±0.3 (0.236±0.012)	1.4±0.1 (0.055±0.004)	4.45 (0.175)	1.6 (0.063)	263
V	7343-20	7.3±0.3 (0.287±0.012)	4.3±0.3 (0.169±0.012)	1.9±0.1 (0.075±0.004)	2.4 (0.094)	1.3 (0.051)	274
H	7360-20	7.3±0.3 (0.287±0.012)	6.0±0.3 (0.236±0.012)	1.9±0.1 (0.075±0.004)	4.45 (0.175)	1.6 (0.063)	385

T527/TF



Case Size		Component Dimensions					Typical Weight
KEMET	EIA	L	W	H	F	S	(mg)
I	3216-10	3.2 ±0.2	1.6 ±0.2	0.9 ±0.1	1.2 ±0.1	0.8 ±0.2	70
08A	3216-08	3.2 ±0.2	1.6 ±0.2	0.8 Maximum	1.2 ±0.1	0.8 ±0.2	
T	3528-12	3.5 ±0.2	2.8 ±0.2	1.1 ±0.1	2.2 ±0.1	0.7 ±0.2	55
S	3216-12	3.2 ±0.2	1.6 ±0.2	1.1 ±0.1	1.2 ±0.1	0.8 ±0.2	26.2

T529

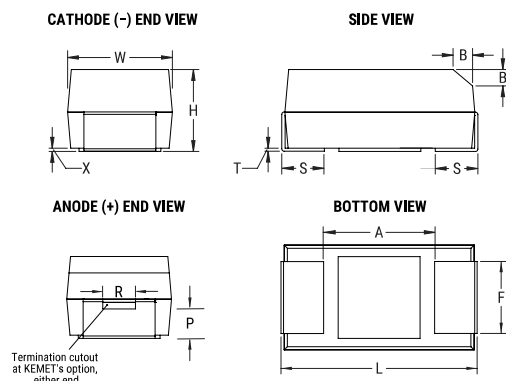


Case Size		Component Dimensions					Typical Weight
KEMET	EIA	L	W	H	F	S	(mg)
P	2012-10	2.0±0.1	1.25±0.1	1.0 maximum	0.9±0.1	0.55±0.1	11.4
I	3216-10	3.2±0.2	1.6±0.2	1.0 maximum	1.2±0.1	0.8±0.1	70

KEMET Part Number: T521B226M025ATE100

For Compal 2020/10/20

T521, Tantalum, Polymer Tantalum, 22 uF, 20%, 25 VDC, SMD, Polymer, Molded, Low ESR, Non-Combustible, 100 mOhms, 3528, Height Max = 2mm, 125°C version



Dimensions

Footprint	3528
L	3.5mm +/-0.2mm
W	2.8mm +/-0.2mm
H	1.9mm +/-0.1mm
T	0.13mm REF
S	0.8mm +/-0.3mm
F	2.2mm +/-0.1mm
A	1.9mm MIN
B	0.4mm +/-0.15mm
P	0.5mm REF
R	1mm REF
X	0.1mm +/-0.1mm

Packaging Specifications

Weight:	94.85 mg
Packaging:	T&R, 178mm
Packaging Quantity:	2000

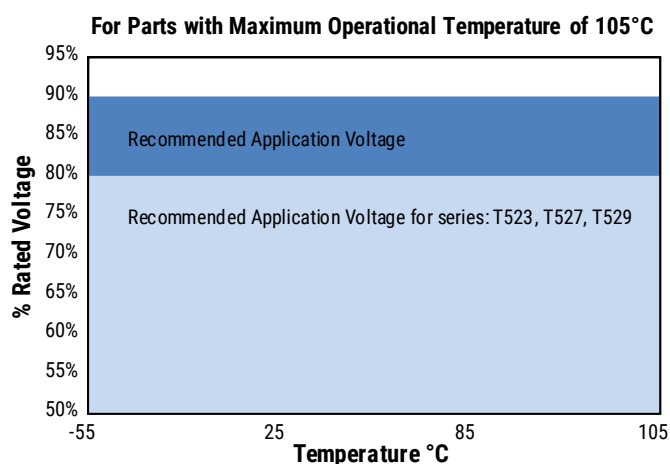
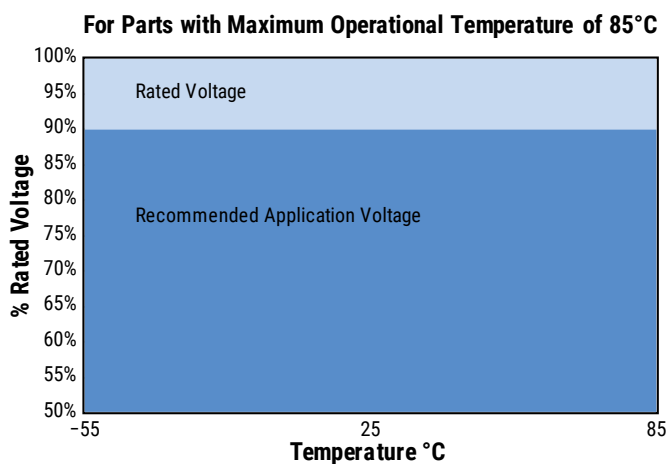
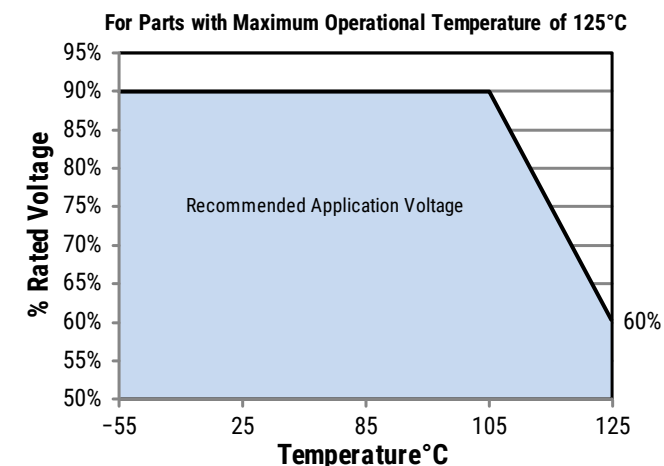
General Information

Series:	T521
Dielectric:	Polymer Tantalum
Style:	SMD Chip
Description:	SMD, Polymer, Molded, Low ESR, Non-Combustible
Features:	Low ESR, High Voltage
RoHS:	Yes
Termination:	Tin
AEC-Q200:	No
Shelf Life:	52 Weeks
MSL:	3

Specifications

Capacitance:	22 uF
Capacitance Tolerance:	20%
Voltage DC:	25 VDC
Temperature Range:	-55/+125°C
Rated Temperature:	125°C
Humidity:	60°C, 90% RH, 500 Hours, No Load
Dissipation Factor:	8% 120Hz 25°C
Failure Rate:	N/A
Resistance:	100 mOhms (100kHz 25°C)
Ripple Current:	1100 mA (100kHz 85°C, delta T=30°C), 770 mA (105°C, delta T=20°C)
Leakage Current:	55 uA (5min 25°C)

Derating Guidelines



Recommended Application Voltage

KO-CAPs are solid state capacitors that demonstrate no wearout mechanism when operated within their recommended guidelines. While the KO-CAP can be operated at full rated voltage, most circuit designers seek a minimum level of assurance in long term reliability, which should be demonstrated with data. A voltage derating can provide the desired level of demonstrated reliability based on industry accepted acceleration models. Since most applications do require long term reliability, KEMET recommends that designers consider a voltage derating, according to the graphic above, for the maximum steady state voltage.

Voltage Rating	Maximum Recommended Steady State Voltage	
	-55°C to 105°C	105°C to 125°C
$75\text{ V} \leq V_R$	90% of V_R	60% of V_R

V_R = Rated Voltage

Ripple Current/Ripple Voltage

Permissible AC ripple voltage and current are related to equivalent series resistance (ESR) and the power dissipation capabilities of the device. Permissible AC ripple voltage, which may be applied is limited by two criteria:

- The positive peak AC voltage plus the DC bias voltage, if any, must not exceed the DC voltage rating of the capacitor.
- The negative peak AC voltage, in combination with bias voltage, if any, must not exceed the allowable limits specified for reverse voltage.

The maximum power dissipation must be reduced with increasing environmental operating temperatures. Refer to the table below for temperature compensation requirements.

Temperature Compensation Multipliers for Maximum Ripple Current		
$T \leq 45^{\circ}\text{C}$	$45^{\circ}\text{C} < T \leq 85^{\circ}\text{C}$	$85^{\circ}\text{C} < T \leq 125^{\circ}\text{C}$
1.00	0.70	0.25

T = Environmental Temperature

Surge Voltage

Surge voltage is the maximum voltage (peak value) which may be applied to the capacitor. The surge voltage must not be applied for periodic charging and discharging in course of normal operation and cannot be part of the application voltage. Surge voltage capability is demonstrated by application of 1,000 cycles at operating temperature. The parts are charged through a 33 Ohm resistor for 30 seconds and then discharged through a 33 Ohm resistor for each cycle.

Rated Voltage (V)	Surge Voltage (V)	Category Voltage (V)	Category Surge Voltage (V)
-55°C to 105°C		up to 125°C	
2.5	3.3	1.7	2.2
6.3	8.2	4.2	5.5
10	13.0	6.7	8.7
16	20.8	10.7	13.9
20	26.0	13.4	17.4
25	32.5	16.8	21.8
35	45.5	23.5	30.5
50	65.0	33.5	43.6

Reverse Voltage

Polymer electrolytic capacitors are polar devices and may be permanently damaged or destroyed if connected in the wrong polarity. These devices will withstand a small degree of transient voltage reversal for short periods as shown in the below table.

Temperature	Permissible Transient Reverse Voltage
25°C	15% of Rated Voltage
55°C	10% of Rated Voltage
85°C	5% of Rated Voltage
105°C	3% of Rated Voltage
125°C*	1% of Rated Voltage

*For series rated to 125°C

Table 2 – Land Dimensions/Courtyard

T520/T521/T525/T530

KEMET	Metric Size Code	Density Level A: Maximum (Most) Land Protrusion (mm)					Density Level B: Median (Nominal) Land Protrusion (mm)					Density Level C: Minimum (Least) Land Protrusion (mm)				
		W	L	S	V1	V2	W	L	S	V1	V2	W	L	S	V1	V2
A	3216-18	1.35	2.20	0.62	6.02	2.80	1.23	1.80	0.82	4.92	2.30	1.13	1.42	0.98	4.06	2.04
B	3528-21	2.35	2.21	0.92	6.32	4.00	2.23	1.80	1.12	5.22	3.50	2.13	1.42	1.28	4.36	3.24
C	6032-28	2.35	2.77	2.37	8.92	4.50	2.23	2.37	2.57	7.82	4.00	2.13	1.99	2.73	6.96	3.74
D	7343-31	2.55	2.77	3.67	10.22	5.60	2.43	2.37	3.87	9.12	5.10	2.33	1.99	4.03	8.26	4.84
H	7360-20	4.25	2.77	3.67	10.22	7.30	4.13	2.37	3.87	9.12	6.80	4.03	1.99	4.03	8.26	6.54
L	6032-19	2.35	2.77	2.37	8.92	4.50	2.23	2.37	2.57	7.82	4.00	2.13	1.99	2.73	6.96	3.74
M	3528-15	2.35	2.20	0.92	6.32	4.00	2.23	1.80	1.12	5.22	3.50	2.13	1.42	1.28	4.36	3.24
Q	7343-12	2.55	2.77	3.67	10.22	5.60	2.43	2.37	3.87	9.12	5.10	2.33	1.99	4.03	8.26	4.84
T	3528-12	2.35	2.20	0.92	6.32	4.00	2.23	1.80	1.12	5.22	3.50	2.13	1.42	1.28	4.36	3.24
U	6032-15	2.35	2.77	2.37	8.92	4.50	2.23	2.37	2.57	7.82	4.00	2.13	1.99	2.73	6.96	3.74
V	7343-21	2.55	2.77	3.67	10.22	5.60	2.43	2.37	3.87	9.12	5.10	2.33	1.99	4.03	8.26	4.84
W	7343-15	2.55	2.77	3.67	10.22	5.60	2.43	2.37	3.87	9.12	5.10	2.33	1.99	4.03	8.26	4.84
X ¹	7343-43	2.55	2.77	3.67	10.22	5.60	2.43	2.37	3.87	9.12	5.10	2.33	1.99	4.03	8.26	4.84
Y ¹	7343-40	2.55	2.77	3.67	10.22	5.60	2.43	2.37	3.87	9.12	5.10	2.33	1.99	4.03	8.26	4.84

Density Level A: For low-density product applications. Recommended for wave solder applications and provides a wider process window for reflow solder processes.

Density Level B: For products with a moderate level of component density. Provides a robust solder attachment condition for reflow solder processes.

Density Level C: For high component density product applications. Before adapting the minimum land pattern variations the user should perform qualification testing based on the conditions outlined in IPC standard 7351 (IPC-7351).

¹ Height of these chips may create problems in wave soldering.

² Land pattern geometry is too small for silkscreen outline.

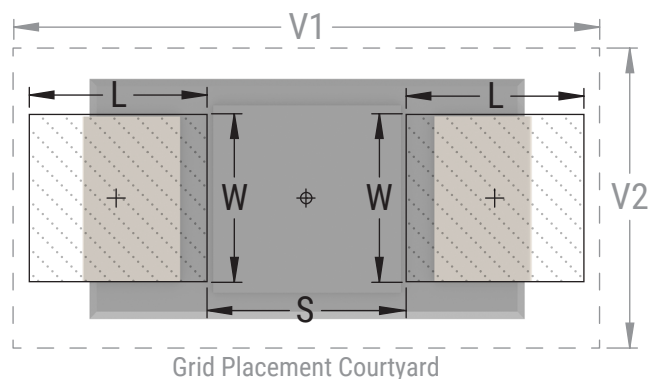
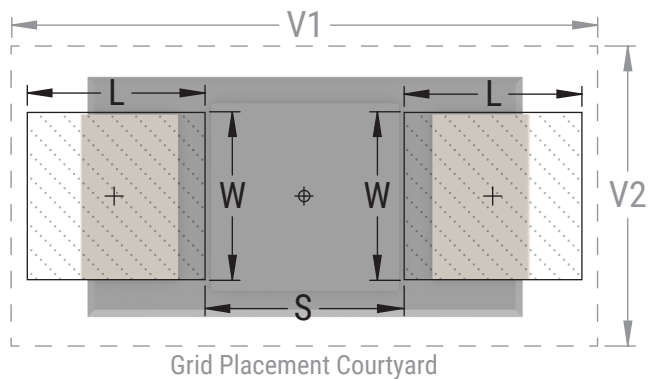


Table 2 – Land Dimensions/Courtyard cont.

T523

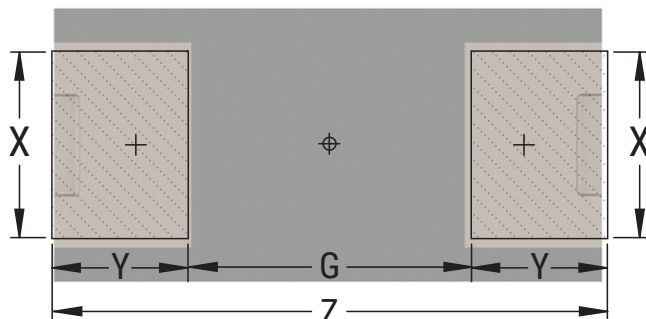
KEMET	Metric Size Code	Maximum (Most) Land Protrusion (mm)					Median (Nominal) Land Protrusion (mm)					Minimum (Least) Land Protrusion (mm)				
Case	EIA	W	L	S	V1	V2	W	L	S	V1	V2	W	L	S	V1	V2
W	7343-15	2.55	2.77	3.67	10.22	5.60	2.43	2.37	3.87	9.12	5.10	2.33	1.99	4.03	8.26	4.84
V	7343-20	2.55	2.77	3.67	10.22	5.60	2.43	2.37	3.87	9.12	5.10	2.33	1.99	4.03	8.26	4.84
G	7360-12	4.60	3.07	3.07	10.22	7.30	4.48	2.67	3.27	9.12	6.80	4.38	2.29	3.43	8.26	6.54
J	7360-15	4.60	3.07	3.07	10.22	7.30	4.48	2.67	3.27	9.12	6.80	4.38	2.29	3.43	8.26	6.54
H	7360-20	4.60	3.07	3.07	10.22	7.30	4.48	2.67	3.27	9.12	6.80	4.38	2.29	3.43	8.26	6.54



T527 / T529

KEMET	Metric Size Code	Dimensions (mm) Minimum – Maximum			
Case	EIA	G	Z	X	Y
A08	3216-08	1.00 – 1.65	3.25 – 3.80	1.1 – 1.30	0.8 – 1.40
I	3216-10	1.00 – 1.65	3.25 – 3.80	1.10 – 1.30	0.80 – 1.40
S	3216-12	1.00 – 1.65	3.25 – 3.80	1.10 – 1.30	0.80 – 1.40
P	2012-10	0.40 – 1.05	2.05 – 2.60	0.80 – 1.00	0.50 – 1.10

(JEITA RC-2371 is recommended for reference)



Soldering Process

The KEMET families of surface mount capacitors are compatible with wave (single or dual), convection, IR, or vapor phase reflow techniques. Preheating of these components is recommended to avoid extreme thermal stress. KEMET's recommended profile conditions for convection and IR reflow reflect the profile conditions of the IPC/J-STD-020D standard for moisture sensitivity testing. The devices can safely withstand a maximum of three reflow passes at these conditions.

Please note that although the X/7343-43 case size can withstand wave soldering, the tall profile (4.3 mm maximum) dictates care in wave process development.

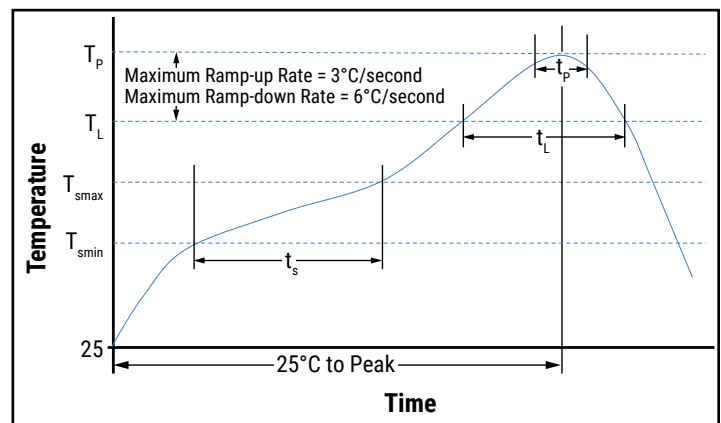
Hand soldering should be performed with care due to the difficulty in process control. If performed, care should be taken to avoid contact of the soldering iron to the molded case. The iron should be used to heat the solder pad, applying solder between the pad and the termination, until reflow occurs. Once reflow occurs, the iron should be removed immediately. "Wiping" the edges of a chip and heating the top surface is not recommended.

Profile Feature	SnPb Assembly	Pb-Free Assembly
Preheat/Soak		
Temperature Minimum (T_{smin})	100°C	150°C
Temperature Maximum (T_{smax})	150°C	200°C
Time (t_s) from T_{smin} to T_{smax}	60 – 120 seconds	60 – 120 seconds
Ramp-up Rate (T_L to T_P)	3°C/seconds maximum	3°C/seconds maximum
Liquidous Temperature (T_L)	183°C	217°C
Time Above Liquidous (t_L)	60 – 150 seconds	60 – 150 seconds
Peak Temperature (T_P)	220°C* 235°C**	250°C* 260°C**
Time within 5°C of Maximum Peak Temperature (t_p)	20 seconds maximum	30 seconds maximum
Ramp-down Rate (T_P to T_L)	6°C/seconds maximum	6°C/seconds maximum
Time 25°C to Peak Temperature	6 minutes maximum	8 minutes maximum

Note: All temperatures refer to the center of the package, measured on the package body surface that is facing up during assembly reflow.

* For Case Size height > 2.5 mm

** For Case Size height ≤ 2.5 mm



Storage

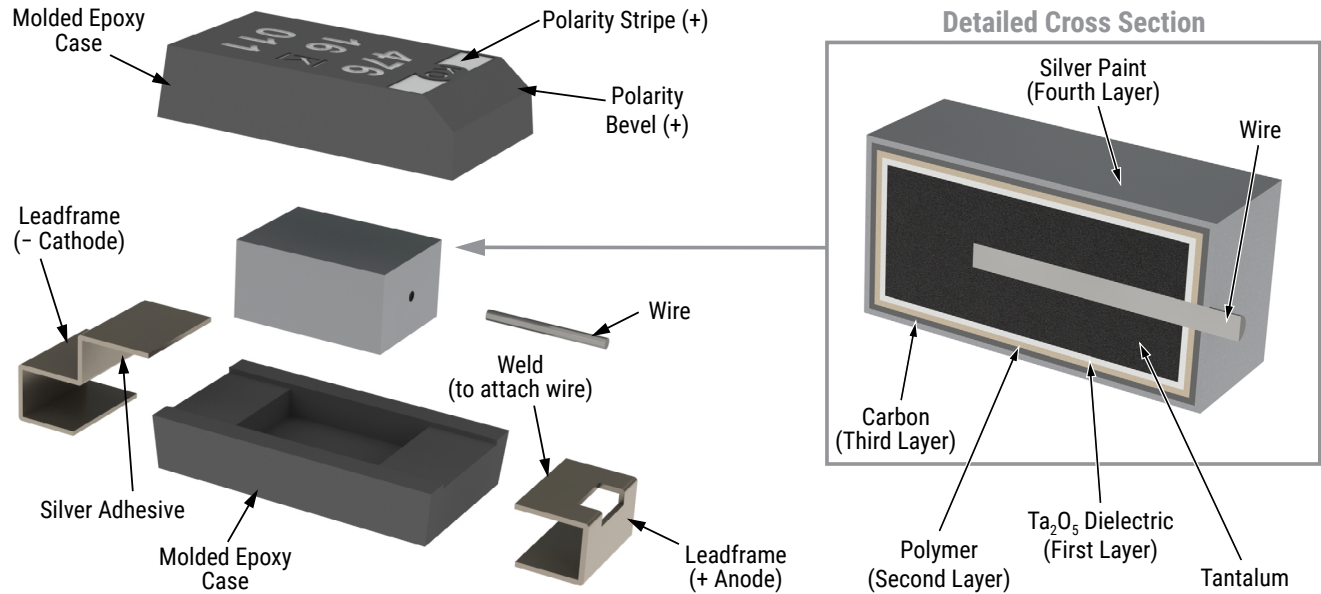
All KO-CAP Series are shipped in moisture barrier bags (MBBs) with desiccant and humidity indicator card (HIC). These parts are classified as MSL3 (Moisture Sensitivity Level 3) per IPC/JEDEC J-STD-020 and packaged per IPC/JEDEC J-STD-033. MSL3 specifies a floor time of 168H at 30°C maximum temperature and 60% relative humidity. Unused capacitors should be sealed in a MBB with fresh desiccant.

The calculated shelf life in a sealed bag would be 12 months from a bag seal date in a storage environment of < 40°C and humidity < 90% RH. It should be 24 months from a bag seal date in a storage environment of < 30°C and humidity < 70% RH.

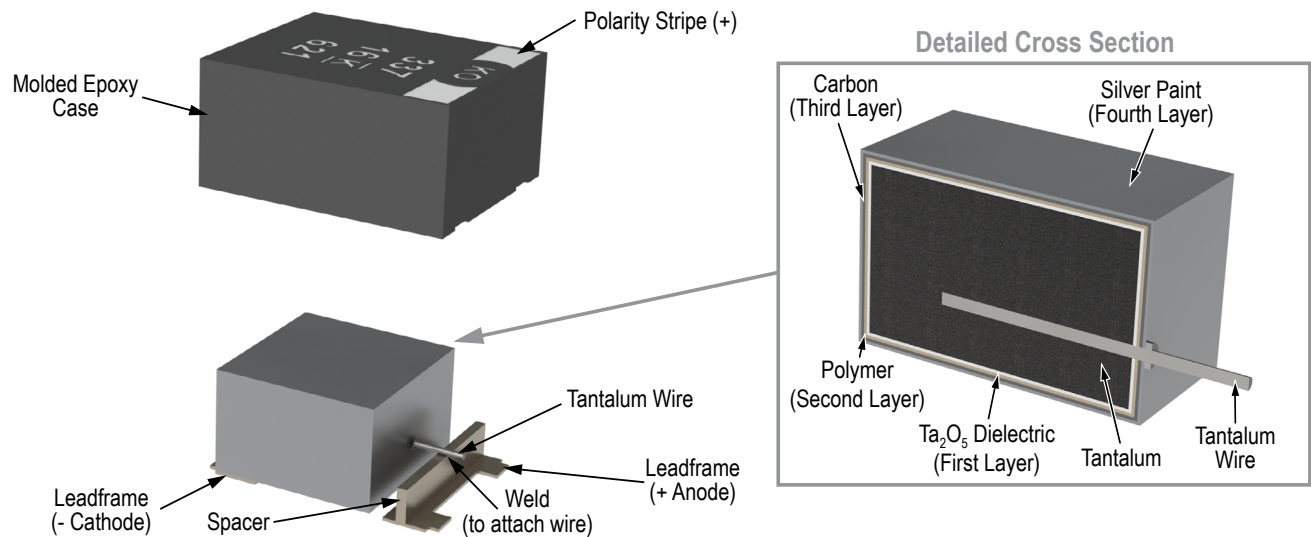
If baking is required, refer to IPC/JEDEC J-STD-033 for bake procedure.

Construction

T520/T521/T525/T530

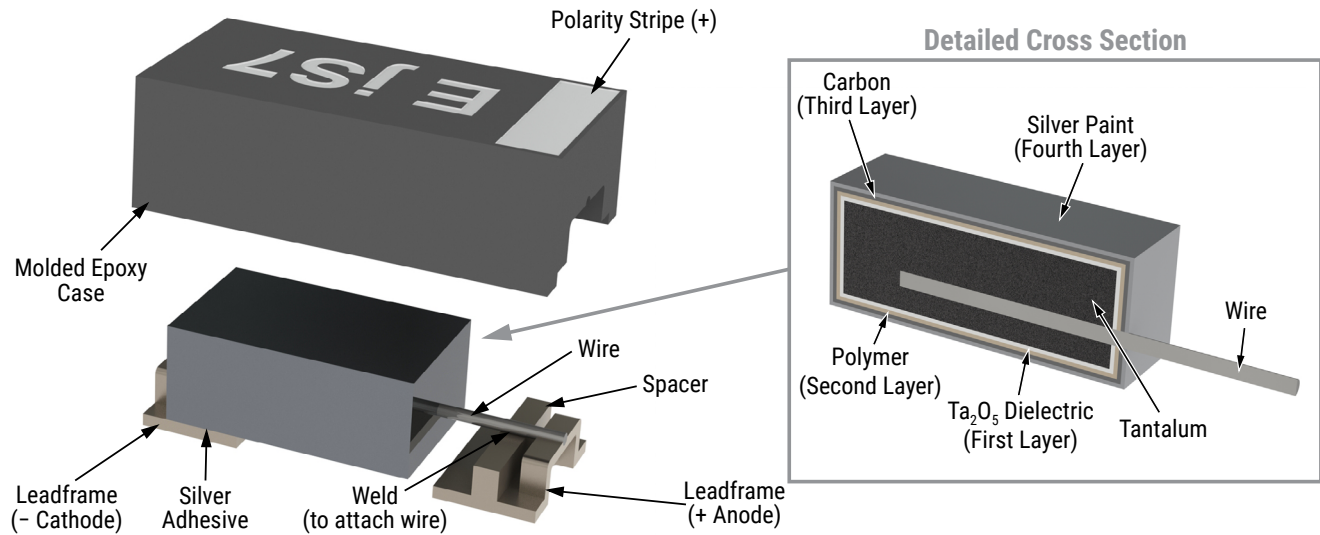


T523

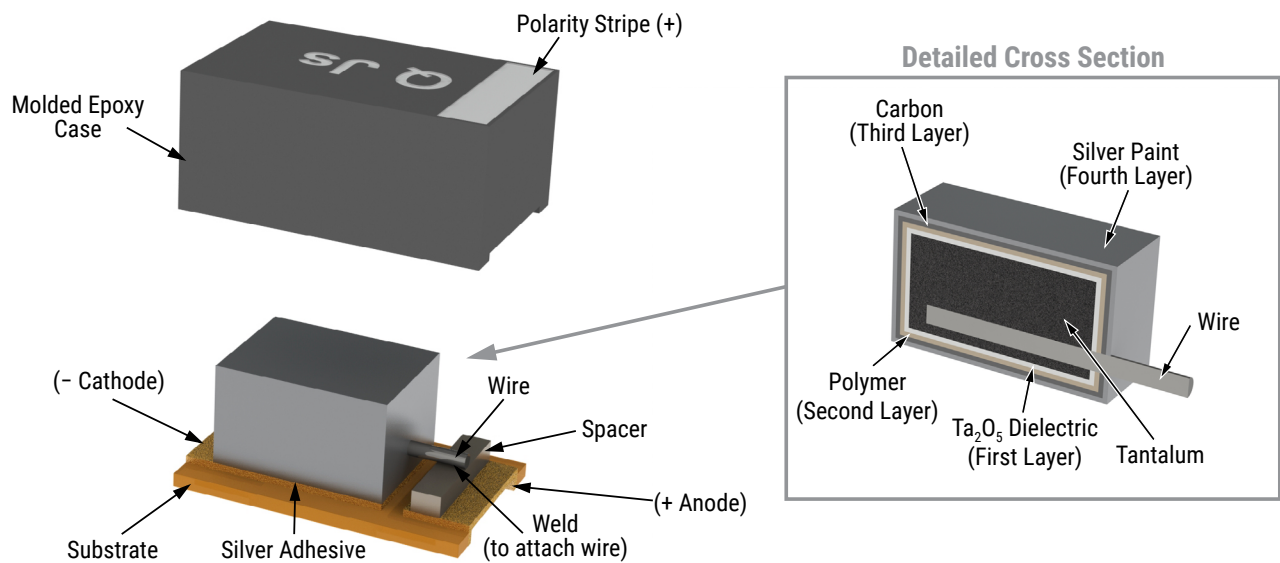


Construction cont.

T527

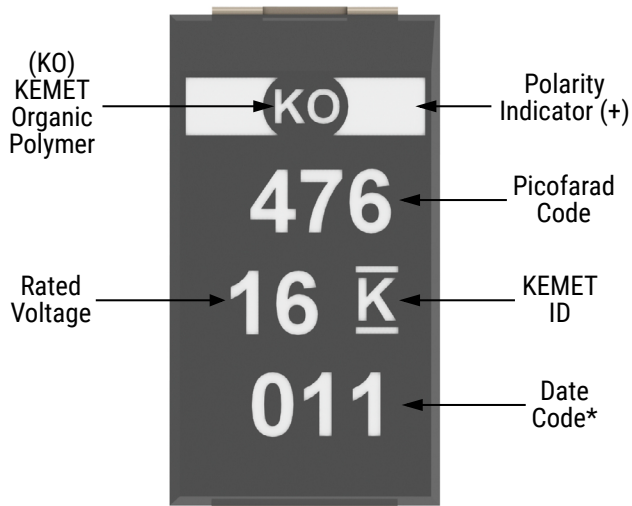


T529



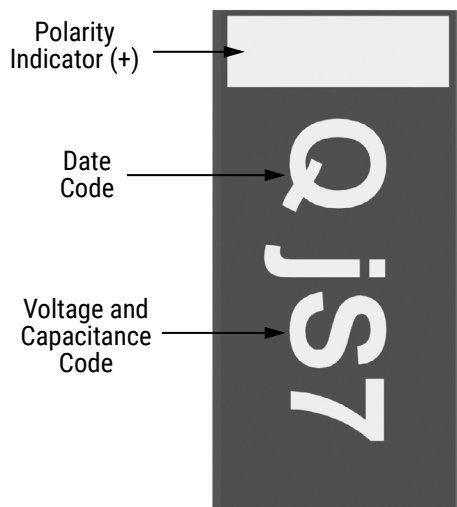
Capacitor Marking

T520/T521/T523/T525/T530



* 011 = 11th week of 2020

Date Code*	
1 st digit = Last number of Year	6 = 2016 7 = 2017 8 = 2018 9 = 2019 0 = 2020
2 nd and 3 rd digit = Week of the Year	01 = 1 st week of the Year to 52 = 52 nd week of the Year



T520A/T527

Code	e	g	J	A	E
Rated Voltage	2.5 V	4 V	6.3 V	10 V	25 V

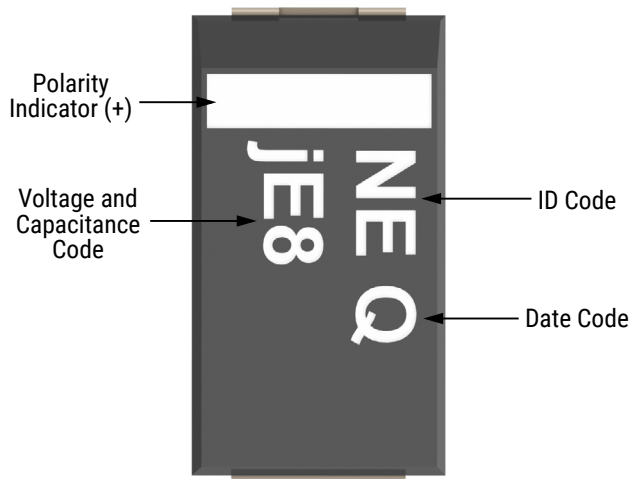
Code	A	E7	J7	N7
Capacitance	10	15	22	33

Code	S7	w	A8	J8
Capacitance	47	68	100	220

Date Code*												
Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2017	A	B	C	D	E	F	G	H	J	K	L	M
2018	N	P	Q	R	S	T	U	V	W	X	Y	Z
2019	a	b	c	d	e	f	g	h	j	k	l	m
2020	n	p	q	r	s	t	u	v	w	x	y	z

Capacitor Marking cont.

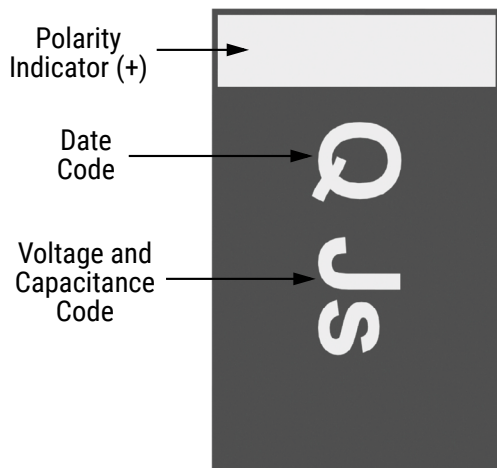
T520B/T521B/T520T



Code	e	g	J	A	C	D	E
Rated Voltage	2.5 V	4 V	6.3 V	10 V	16 V	20 V	25 V

Code	E7	J7	N7	A8
Capacitance	15	22	33	100

Code	E8	J8	N8	S8
Capacitance	150	220	330	470



T529

Code	J	A	C	E
Rated Voltage	6.3 V	10 V	16 V	25 V

Code	S	a	j	s
Capacitance (μF)	4.7	10	22	47

Date Code*												
Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2017	A	B	C	D	E	F	G	H	J	K	L	M
2018	N	P	Q	R	S	T	U	V	W	X	Y	Z
2019	a	b	c	d	e	f	g	h	j	k	l	m
2020	n	p	q	r	s	t	u	v	w	x	y	z

Tape & Reel Packaging Information

KEMET's molded chip capacitor families are packaged in 8 and 12 mm plastic tape on 7" and 13" reels in accordance with *EIA Standard 481: Embossed Carrier Taping of Surface Mount Components for Automatic Handling*. This packaging system is compatible with all tape-fed automatic pick-and-place systems.

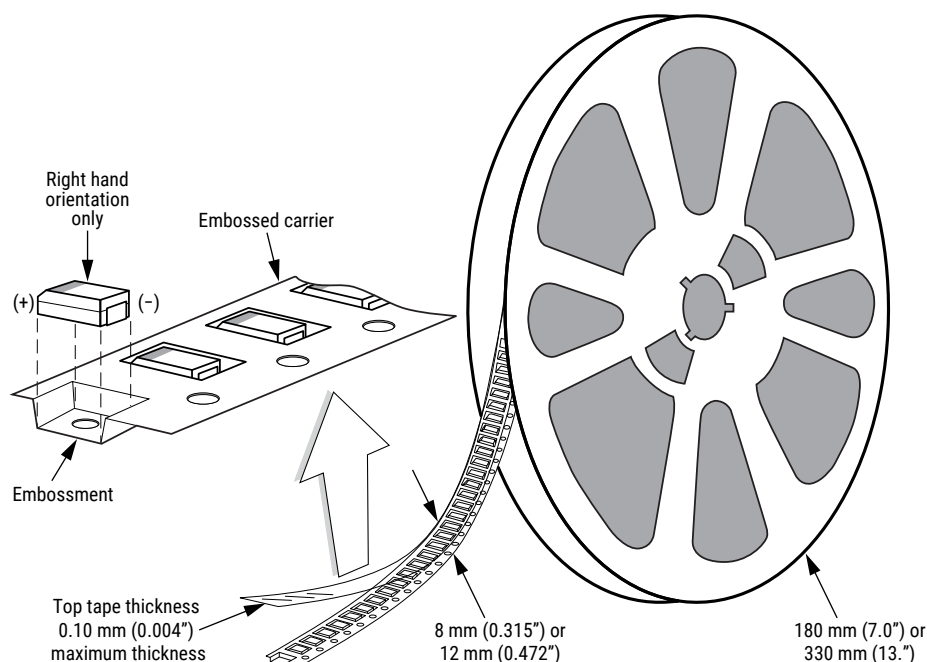


Table 3 – Packaging Quantity

Case Code		Tape Width (mm)	7" Reel*	13" Reel*
KEMET	EIA			
P	2012-10	8	3,000	N/A
R	2012-12	8	2,500	10,000
08A	3216-08	8	3,000	N/A
I	3216-10	8	3,000	N/A
S	3216-12	8	3,000	N/A
A	3216-18	8	2,000	N/A
T	3528-12	8	3,000	10,000
M	3528-15	8	2,500	8,000
B	3528-21	8	2,000	8,000
U	6032-15	12	1,000	5,000
L	6032-19	12	1,000	3,000
C	6032-28	12	500	3,000
Q	7343-12	12	1,000	3,000
W	7343-15	12	1,000	3,000
Z	7343-17	12	1,000	3,000
V	7343-19	12	1,000	3,000
D	7343-31	12	500	2,500
Y	7343-40	12	500	2,000
X	7343-43	12	500	2,000
J	7360-15	12	1,000	3,000
H	7360-20	12	1,000	3,000
O	7360-43	12	250	1,000

* No C-Spec required for 7" reel packaging. C-7280 required for 13" reel packaging.

Figure 1 – Embossed (Plastic) Carrier Tape Dimensions

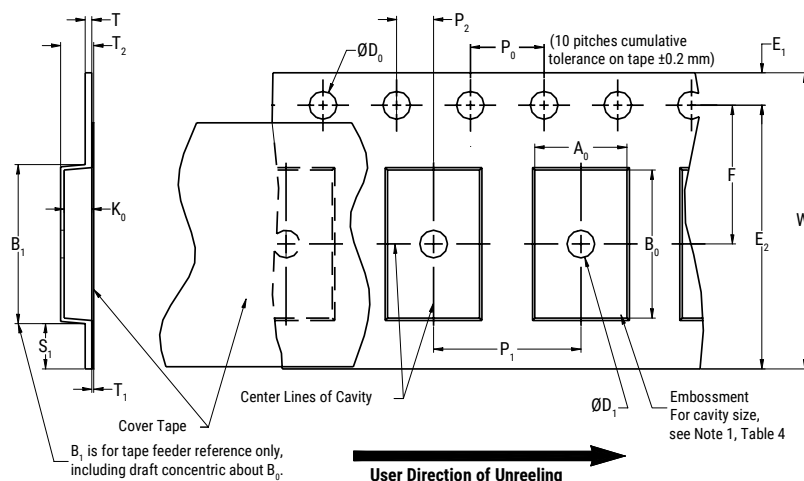


Table 4 – Embossed (Plastic) Carrier Tape Dimensions

Metric will govern

Constant Dimensions – Millimeters (Inches)									
Tape Size	D ₀	D ₁ Minimum Note 1	E ₁	P ₀	P ₂	R Reference Note 2	S ₁ Minimum Note 3	T Maximum	T ₁ Maximum
8 mm	1.5 +0.10/-0.0 (0.059 +0.004/-0.0)	1.0 (0.039)	1.75 ±0.10 (0.069 ±0.004)	4.0 ±0.10 (0.157 ±0.004)	2.0 ±0.05 (0.079 ±0.002)	25.0 (0.984)	0.600 (0.024)	0.600 (0.024)	0.100 (0.004)
12 mm		1.5 (0.059)				30 (1.181)			

Variable Dimensions – Millimeters (Inches)								
Tape Size	Pitch	B ₁ Maximum Note 4	E ₂ Minimum	F	P ₁	T ₂ Maximum	W Maximum	A ₀ , B ₀ & K ₀
8 mm	Single (4 mm)	4.35 (0.171)	6.25 (0.246)	3.5 ±0.05 (0.138 ±0.002)	2.0 ±0.05 or 4.0 ±0.10 (0.079 ±0.002 or 0.157 ±0.004)	2.5 (0.098)	8.3 (0.327)	Note 5
12 mm	Single (4 mm) and Double (8 mm)	8.2 (0.323)	10.25 (0.404)	5.5 ±0.05 (0.217 ±0.002)	2.0 ±0.05 (0.079 ±0.002) or 4.0 ±0.10 (0.157 ±0.004) or 8.0 ±0.10 (0.315 ±0.004)	4.6 (0.181)	12.3 (0.484)	

1. The embossment hole location shall be measured from the sprocket hole controlling the location of the embossment. Dimensions of embossment location and hole location shall be applied independent of each other.
2. The tape, with or without components, shall pass around R without damage (see Figure 4).
3. If $S_1 < 1.0$ mm, there may not be enough area for cover tape to be properly applied (see EIA Standard 481-D, paragraph 4.3, section b).
4. B_1 dimension is a reference dimension for tape feeder clearance only.
5. The cavity defined by A_0 , B_0 and K_0 shall surround the component with sufficient clearance that:
 - (a) the component does not protrude above the top surface of the carrier tape.
 - (b) the component can be removed from the cavity in a vertical direction without mechanical restriction, after the top cover tape has been removed.
 - (c) rotation of the component is limited to 20° maximum for 8 and 12 mm tapes (see Figure 2).
 - (d) lateral movement of the component is restricted to 0.5 mm maximum for 8 mm and 12 mm wide tape (see Figure 3).
 - (e) see Addendum in EIA Standard 481-D for standards relating to more precise taping requirements.

Packaging Information Performance Notes

- 1. Cover Tape Break Force:** 1.0 Kg minimum.
- 2. Cover Tape Peel Strength:** The total peel strength of the cover tape from the carrier tape shall be:

Tape Width	Peel Strength
8 mm	0.1 to 1.0 Newton (10 to 100 gf)
12 mm	0.1 to 1.3 Newton (10 to 130 gf)

The direction of the pull shall be opposite the direction of the carrier tape travel. The pull angle of the carrier tape shall be 165° to 180° from the plane of the carrier tape. During peeling, the carrier and/or cover tape shall be pulled at a velocity of 300 ±10 mm/minute.

- 3. Labeling:** Bar code labeling (standard or custom) shall be on the side of the reel opposite the sprocket holes. Refer to EIA Standards 556 and 624.

Figure 2 – Maximum Component Rotation

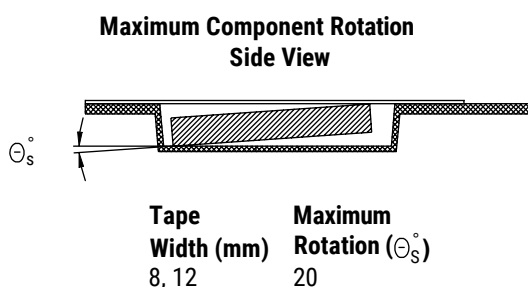
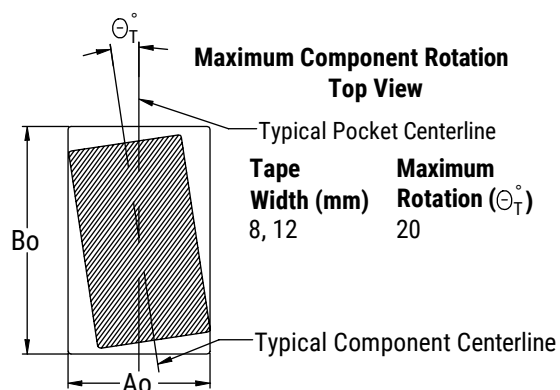


Figure 3 – Maximum Lateral Movement

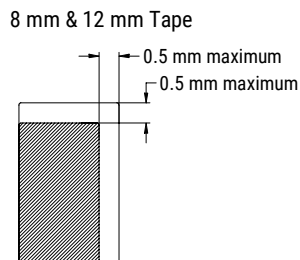


Figure 4 – Bending Radius

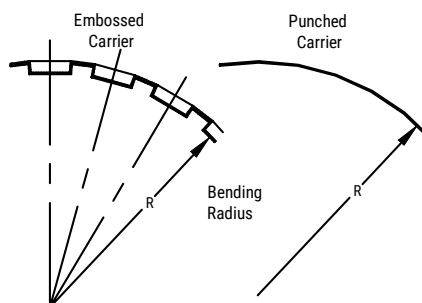
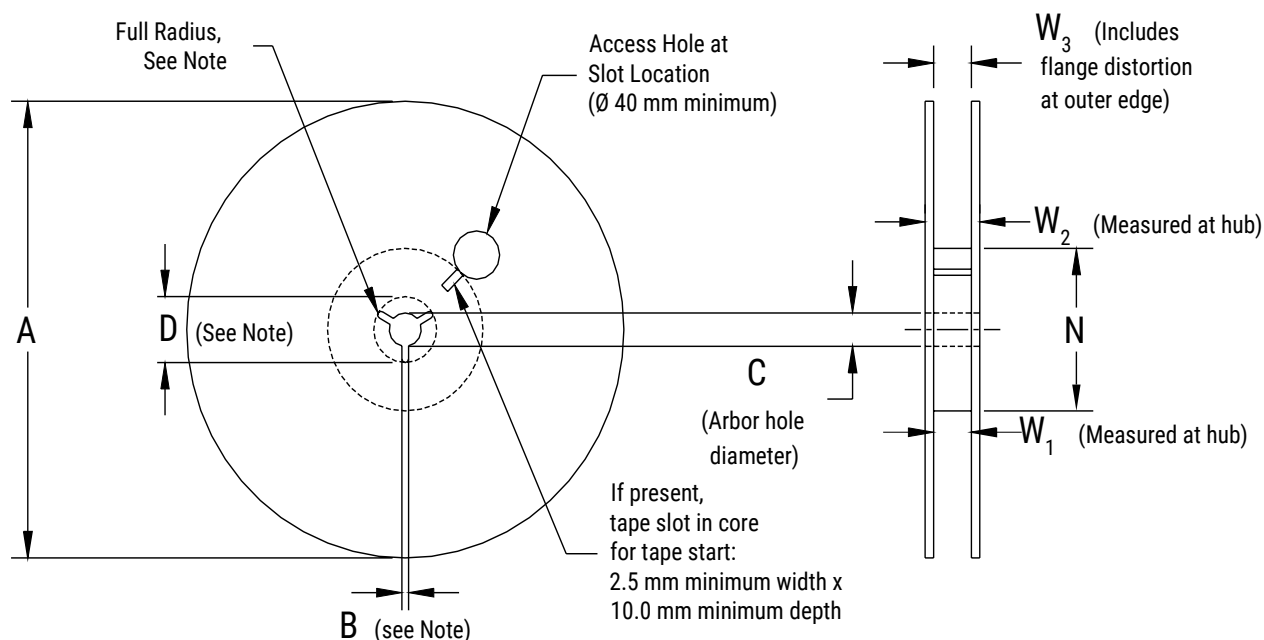


Figure 5 – Reel Dimensions



Note: Drive spokes optional; if used, dimensions B and D shall apply.

Table 5 – Reel Dimensions

Metric will govern

Constant Dimensions – Millimeters (Inches)				
Tape Size	A	B Minimum	C	D Minimum
8 mm	178 ±0.20 (7.008 ±0.008) or 330 ±0.20 (13.000 ±0.008)	1.5 (0.059)	13.0 +0.5/-0.2 (0.521 +0.02/-0.008)	20.2 (0.795)
12 mm				
Variable Dimensions – Millimeters (Inches)				
Tape Size	N Minimum	W ₁	W ₂ Maximum	W ₃
8 mm	50 (1.969)	8.4 +1.5/-0.0 (0.331 +0.059/-0.0)	14.4 (0.567)	Shall accommodate tape width without interference
12 mm		12.4 +2.0/-0.0 (0.488 +0.078/-0.0)	18.4 (0.724)	

Figure 6 – Tape Leader & Trailer Dimensions

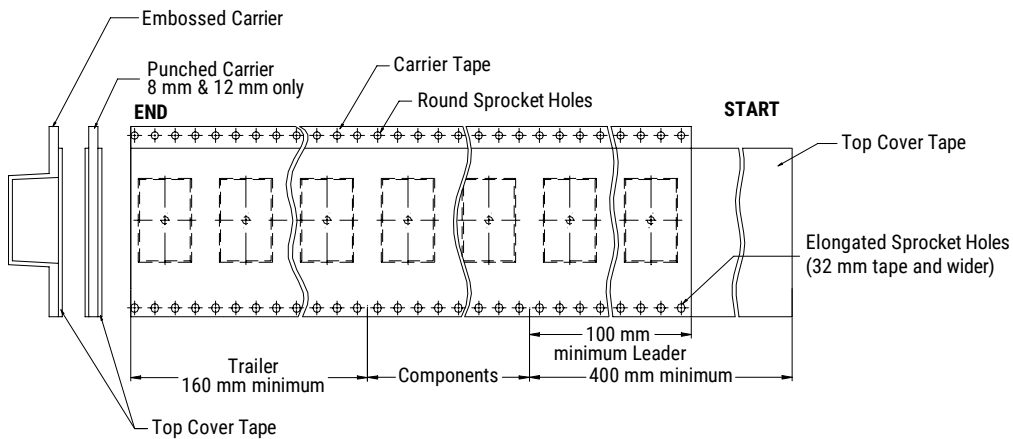
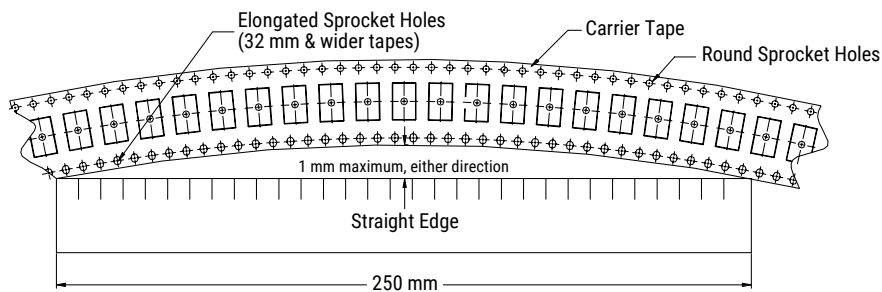


Figure 7 – Maximum Camber



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